Zadanie 27

Oblicz granicę $\lim_{n\to\infty}\frac{2^n+7\cdot 3^{n-1}}{2^n+2\cdot 3^n}$.

① Obliczamy granicę.

$$\lim_{n\to\infty}\frac{2^n+7\cdot 3^{n-1}}{2^n+2\cdot 3^n}=\lim_{n\to\infty}\frac{2^n+7\cdot\frac{1}{3}\cdot 3^n}{2^n+2\cdot 3^n}=\lim_{n\to\infty}\frac{\cancel{3^n}\left(\left(\frac{2^n}{3^n}\right)^{\to 0}+\frac{7}{3}\right)}{\cancel{3^n}\left(\left(\frac{2^n}{3^n}\right)_{\to 0}+2\right)}=\frac{\frac{7}{3}}{2}=\frac{7}{6}$$

Odp.: $\lim_{n\to\infty}\frac{2^n+7\cdot 3^{n-1}}{2^n+2\cdot 3^n}=\frac{7}{6}$.